

Literature Status for Higher-Rank Numerical Ranges

arXiv:math/0511278 answered by arXiv:0706.1536

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Status. `literature_already_answered`. This note records an exact later theorem, not a new proof of the run.

Source questions

For $A \in M_n(\mathbb{C})$ and $1 \leq k \leq n$, Choi–Kribs–Życzkowski define

$$\Lambda_k(A) = \{\mu \in \mathbb{C} : PAP = \mu P \text{ for some rank-}k \text{ orthogonal projection } P\}.$$

Conjecture 2.8 of arXiv:math/0511278 asks whether, for a normal matrix with eigenvalues $\lambda_1, \dots, \lambda_n$ counted with multiplicity,

$$\Lambda_k(A) = \bigcap_{1 \leq j_1 < \dots < j_{n-k+1} \leq n} \text{conv}\{\lambda_{j_1}, \dots, \lambda_{j_{n-k+1}}\}. \quad (\text{CK}\acute{\text{Z}})$$

Problem 2.9 asks more generally whether $\Lambda_k(A)$ is convex whenever it is nonempty. The paper identifies the first unresolved normal case as $(n, k) = (5, 2)$ and asks whether all interior points of the pentagonal region for the cyclic five-shift belong to Λ_2 .

Complete later answer

Li–Sze, arXiv:0706.1536, Theorem 2.2, prove for every complex matrix A that

$$\Lambda_k(A) = \bigcap_{\xi \in [0, 2\pi)} \left\{ \mu \in \mathbb{C} : e^{i\xi} \mu + e^{-i\xi} \bar{\mu} \leq \lambda_k \left(e^{i\xi} A + e^{-i\xi} A^* \right) \right\}, \quad (\text{LS})$$

where $\lambda_k(H)$ denotes the k th largest eigenvalue of a Hermitian matrix H . Every set in the intersection is a closed half-plane. Corollary 2.3 therefore gives

$\Lambda_k(A) \text{ is convex for every } A \in M_n(\mathbb{C}).$

This answers Problem 2.9 affirmatively, including the empty-set convention and hence slightly strengthening the source’s phrasing.

For normal A , Li–Sze Corollary 2.4 reduces formula (LS) to formula (CK $\acute{\text{Z}}$) verbatim. Thus Conjecture 2.8 is also completely affirmative. In particular, the cyclic five-shift’s Λ_2 is the full intersection of the five four-eigenvalue convex hulls: every interior point of the inner pentagon is included.

Identification boundary

The intermediate paper arXiv:quant-ph/0608244 verified the conjecture for a wide class of unitary and normal matrices. Li–Sze’s Theorem 2.2 and Corollaries 2.3–2.4 are the decisive general result: they cover arbitrary matrices for convexity and every finite normal matrix for the exact spectral formula. No additional hypothesis or parameter translation is needed.

References

- [1] M.-D. Choi, D. W. Kribs, and K. Życzkowski, *Higher-Rank Numerical Ranges and Compression Problems*, arXiv:math/0511278; Linear Algebra Appl. 418 (2006), 828–839.
- [2] C.-K. Li and N.-S. Sze, *Canonical forms, higher rank numerical range, convexity, totally isotropic subspace, matrix equations*, arXiv:0706.1536; Proc. Amer. Math. Soc. 136 (2008), 3013–3023, Theorem 2.2 and Corollaries 2.3–2.4.